

BLDC FOC With Selectable Position Sensor

This example runs the same cascaded control structure with any of the position sensors already wired on the Ownverter shield:

- Hall
- ABZ incremental encoder
- Sin/Cos analog sensor

The sensor selection is done entirely from the devicetree overlay. `src/main.cpp` now reads the active sensor through `PositionAPI` and only uses ABZ-specific count data when the selected sensor is actually ABZ.

Files

- `src/hall.overlay`: turnkey Hall configuration
- `src/abz.overlay`: turnkey ABZ configuration
- `src/sincos.overlay`: turnkey Sin/Cos configuration
- `src/app.overlay`: active overlay used by the build

How To Select The Sensor

1. Copy one of the provided overlays onto `src/app.overlay`.
2. Adjust motor and sensor parameters if needed.
3. Build and flash.

Example:

```
cp src/abz.overlay src/app.overlay
```

Overlay Content

Each overlay:

- selects the sensor through `/chosen/owntech,position-sensor`
- sets `&default_motor { pole-pairs = <...>; }`
- overrides the sensor-specific parameters on the shield node already declared by `ownverter_v1_1_0.overlay`

The hardware pin mapping remains the one defined by the shield:

- Hall: spin header pins 7, 9, 49
- ABZ: timer `timers3`
- Sin/Cos: `ANALOG_SIN` on pin 43, `ANALOG_COS` on pin 45

Parameters To Check

Before running the motor, verify these values in the chosen overlay:

- `pole-pairs`
- `direction-sign`
- `electrical-offset`
- `counts-per-revolution` for ABZ
- `hall-sector-table` and `hall-interpolation` for Hall

`electrical-offset` is stored as the raw IEEE754 `float32` bit pattern expected by the binding. `0x00000000` means `0.0f`.

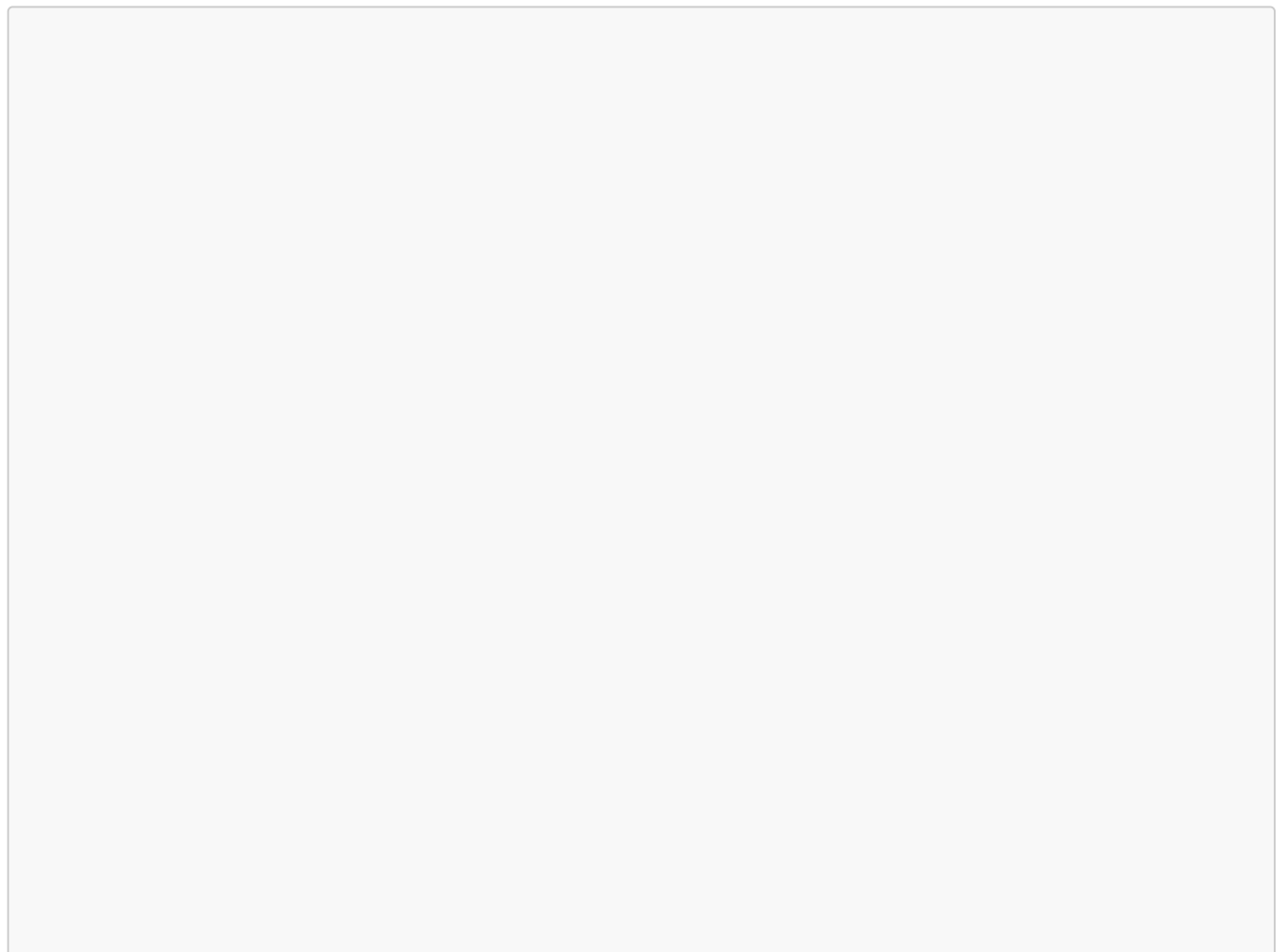
Runtime Behavior

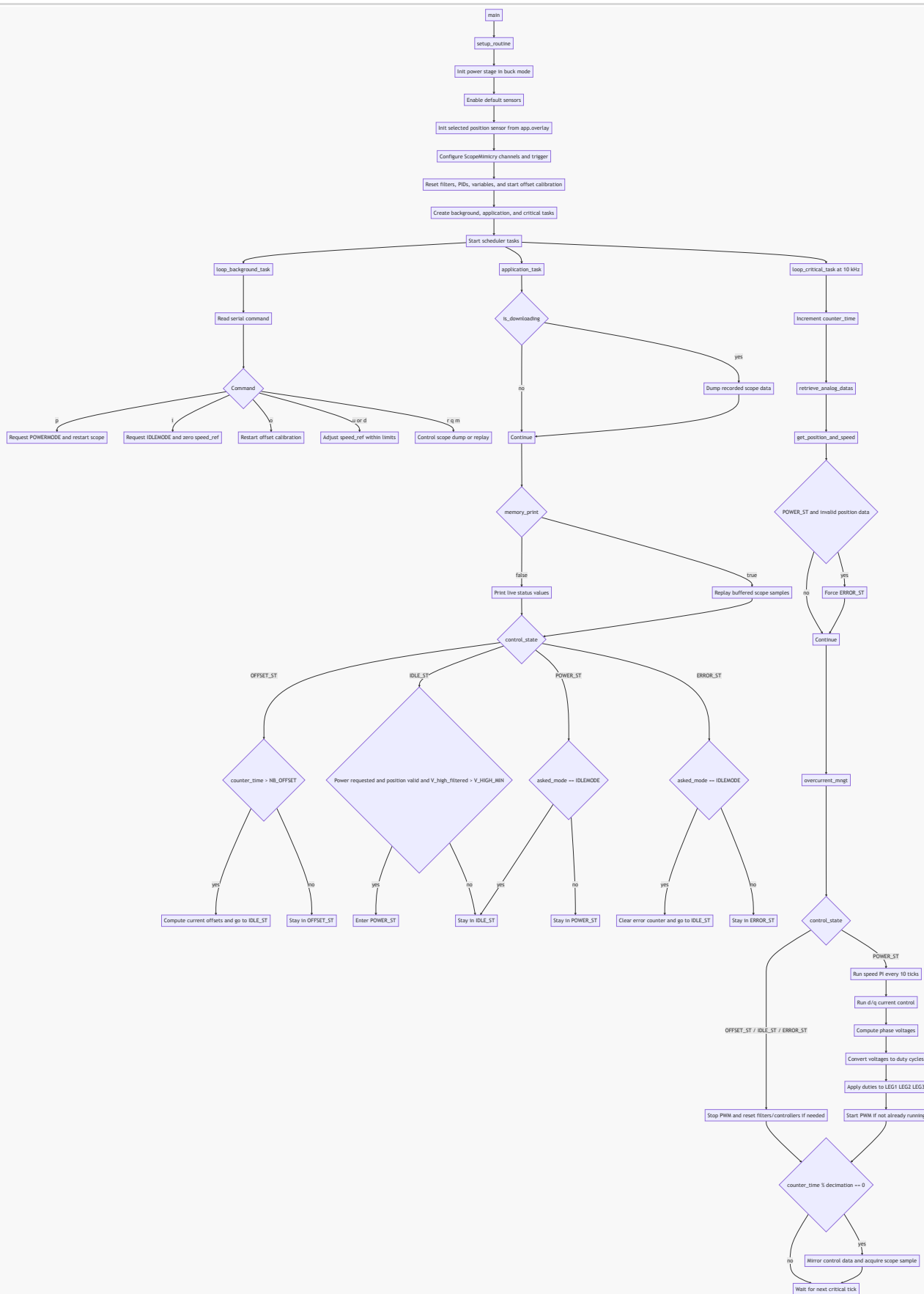
`main.cpp` uses `shield.position.initDefault()` and expects the selected overlay to define the active sensor. The control only enters `POWER_ST` when:

- the position sensor initialized correctly
- the position update is valid
- the DC bus voltage is above the startup threshold

If position feedback becomes invalid while running, the control falls back to `ERROR_ST` instead of continuing with stale rotor angle data.

Control Flow





Serial Commands

- **p**: request power mode

- **i**: request idle mode
- **o**: restart current offset calibration
- **u**: increase speed reference
- **d**: decrease speed reference
- **r**: dump scope data
- **q**: restart scope acquisition
- **m**: toggle buffered scope replay

Hall Notes

The default Hall sector table is:

```
{5, 1, 0, 3, 4, 2}
```

This maps raw Hall states **001** to **110** to electrical sectors. If your motor phases or Hall wiring are permuted, this table or **direction-sign** may need adjustment.

Phase Mapping

Electrical phase	PWM	LEG
Phase A	PWMA	LEG1
Phase B	PWMC	LEG2
Phase C	PWME	LEG3